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Revisiting Algorithmic Bounded Rationality With Potential Solutions: How Recommendation Systems Amplify Cognitive Biases in Consumer Decision-Making

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ABSTRACT

Algorithm-driven platforms have become central to modern consumer decision-making. While consumers have always been subject to bounded rationality, recent discussion suggests that personalized recommendation systems can further constrain consumer choices and, in turn, reinforce various cognitive biases. This article explores the theoretical underpinnings of bounded rationality in the context of algorithmically curated content, outlines key biases that are exacerbated by these systems, and discusses the societal implications of algorithmic confinement—from political polarization to consumer rights concerns. Finally, it proposes potential solutions, including policy-based interventions, forced diversity mechanisms, and consumer education approaches that can mitigate the adverse effects of algorithm-driven recommendation systems.

1 | Introduction

Consumer choice has never been limitless. From Simon's (1955) concept of bounded rationality, it is widely recognized that individuals face cognitive, informational, and temporal constraints in decision-making. Bounded rationality provides the constraint set, while cognitive biases are heuristic by-products that can be amplified under algorithmic curation. These limitations are exacerbated by algorithms that shape interactions across various platforms, including e-commerce and social media feeds (Noble 2018). Recommendation systems across commerce and media leverage user data to tailor content efficiently. While these personalized recommendations offer convenience by simplifying complex choices, empirical evidence suggests that they can also limit consumer options, reinforcing cognitive biases such as confirmation bias and anchoring. Such constraints limit consumer exploration and may exacerbate societal concerns, such as ideological polarization and reduced market diversity (Adomavicius et al. 2013; Heo et al. 2021; Sunstein 2009). While platforms like Amazon diversify recommendations, thereby avoiding extreme filter bubbles (Pariser 2011), recent studies reveal confinement to niche categories or ideological content (Bandy 2021; Urman et al. 2024). This commentary synthesizes recent empirical evidence, including Amazon pricing examinations (Angwin and Mattu 2016) and Facebook field experiments (Guess et al. 2023), to reframe algorithmic bounded rationality across diverse contexts. It examines bias amplification and proposes tailored solutions. Beyond platform-controlled recommenders, a complementary path is the use of user-owned, low-cost recommendation agents that re-rank platform outputs or operate independently, shifting optimization from firm profits toward user welfare and autonomy.

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2 | Bounded Rationality in the Algorithmic Age

2.1 | Classic Bounded Rationality

Simon (1955) introduced bounded rationality, emphasizing that individuals do not always behave as perfectly rational agents. Instead, people rely on heuristics due to cognitive constraints, limited information, and time pressures. However, traditional bounded rationality assumes constraints primarily arise from human cognitive capacity or available information. The rise of big data and machine learning-based recommendations adds an external constraint: algorithm-driven choices (Hosanagar 2019).

2.2 | Algorithmic Bounded Rationality

Algorithmic bounded rationality occurs when recommendation systems create additional external constraints, further narrowing consumer choices beyond inherent cognitive limitations. Empirical evidence shows that repeated algorithmic recommendations, such as Amazon's search results, increasingly limit product diversity by reinforcing past behaviors and preferences, even with broader search inputs (Heo et al. 2021; Linden et al. 2003). Rather than merely reflecting existing preferences, algorithms actively amplify and reinforce them, steering users toward familiar options and restricting exposure to diverse alternatives (Pariser 2011).

2.3 | Accumulated Empirical Evidence on Algorithmic Confinement

Empirical studies confirm that recommender systems substantially influence consumer decision-making by simplifying complex choice sets and reducing cognitive effort, leading consumers to heavily rely on algorithmically suggested options (Chen et al. 2013). This phenomenon, termed "algorithm overdependence," reflects consumers perceiving recommendations as implicitly validated (Banker and Khetani 2019). Investigations further demonstrate that users consistently choose prominently positioned recommendations, despite potentially superior alternatives (Bandy 2021), and Amazon's recommendations anchor consumers to narrower, familiar product subsets (Kumar and Hosanagar 2019). These findings empirically validate algorithmic bounded rationality as a real-world phenomenon shaping consumer decisions.

3 | Amplification of Cognitive Biases

3.1 | Confirmation Bias

Confirmation bias refers to individuals' tendency to seek, favor, and recall information that aligns with their preexisting beliefs while discounting contradictory evidence (Nickerson 1998). Driven partly by motivated reasoning—where individuals reinforce attitudes to reduce cognitive dissonance (Kunda 1990)—this bias often results in self-selected echo chambers, such as social circles or media sources (Sunstein 2009). Algorithmic recommendation systems substantially amplify confirmation bias by strategically suggesting content aligned with prior behaviors (Covington et al. 2016). For example, YouTube users repeatedly viewing politically conservative videos receive increasingly similar recommendations, limiting their exposure to alternative perspectives and reinforcing ideological echo chambers (Haroon et al. 2023). Similarly, recommendation systems steer shoppers repeatedly choosing eco-friendly items toward similar "green" products, restricting their exploration of diverse alternatives.

3.2 | Anchoring Effects

The anchoring effect describes individuals' reliance on initial reference points ("anchors") when making subsequent judgments, even if these anchors are arbitrary or irrelevant (Tversky and Kahneman 1974). Anchoring occurs because people typically fail to sufficiently adjust from these initial references, leading to systematic biases in valuations and decisions (Furnham and Boo 2011). Adomavicius et al. (2013) demonstrated that recommender systems act as powerful anchors, establishing implicit benchmarks through initial recommendations. On Amazon, products and price brackets that are prominently recommended become implicit comparison standards, constraining consumers' perceptions of acceptable prices and reducing their product diversity. Consequently, consumers often accept narrower product categories and higher price ranges, illustrating tangible economic impacts (Hannak et al. 2014). Rather than treating platform labels and badges as a distinct "framing" construct, the manuscript treats them as anchoring and heuristic salience mechanisms that shift reference points and attention, thereby shaping valuations and choice sets (Tversky and Kahneman 1974; Adomavicius et al. 2013).

4 | Societal Implications of Algorithmic Confinement

Numerous studies have linked social media algorithms to intensified political polarization, as recommendation systems reinforce existing beliefs and foster insular communities (Pariser 2011; Sunstein 2009). The 'filter bubble' metaphor has been widely critiqued for overreach and for limited, context-dependent empirical support (Bruns 2019; Bruns 2024). Evidence suggests heterogeneity, as average users often encounter more diversity than initially presumed, with stronger enclosure concentrated among highly engaged or ideologically driven cohorts. This is why the discussion here emphasizes broader consumer welfare and market structure implications beyond political content. While filter bubbles may not directly cause extremism, they amplify it by limiting exposure to diverse perspectives. Algorithmic bounded rationality also raises concerns regarding consumer autonomy, fairness, and informed choice, as algorithms prioritizing profitability and engagement can stifle competition, marginalize niche products, and restrict consumers' exposure to valuable alternatives (O'Neil 2016; Schafer et al. 2001). Increasing regulatory scrutiny focuses on algorithmic transparency and bias, advocating for disclosures about recommender system functioning (Brauneis and Goodman 2018). However, effective oversight

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remains challenging due to the systems' technical complexity and proprietary nature.

Although it is beyond the scope of this commentary to exhaustively explore all dimensions of recommendation biases, the three additional considerations below warrant further elaboration to deepen our understanding of algorithmic bounded rationality and associated cognitive biases.

4.1 | Product Complexity and Consumer Expertise

Recommender systems influence consumer decisions when individuals lack familiarity or face excessive complexity in product categories. Consumer expertise critically shapes how effectively individuals evaluate recommended items (Alba and Hutchinson 1987). In scenarios where consumers have limited expertise or find choices excessively complex, algorithmically curated recommendations simplify the choice set and significantly reduce cognitive effort (Coupey et al. 1998; Dellaert and Stremersch 2005). However, reliance on algorithmic simplifications in such contexts may inadvertently foster "pseudo-expertise," where consumers incorrectly perceive themselves as well informed, potentially overlooking superior but unfamiliar alternatives. This scenario underscores a nuanced challenge: recommender systems can aid decision-making under complexity but may also lead to excessive consumer passivity and insufficient exploration of optimal choices.

4.2 | Anchoring and Profit-Driven Bias

A significant policy-related concern regarding recommender systems is their tendency to anchor consumers to options that maximize company profitability rather than consumer utility (Angwin and Mattu 2016; Hannak et al. 2014). Such profit-driven anchoring is empirically documented in audits of major online retailers, where algorithms often steer consumers toward higher-priced or higher-margin items, subtly excluding economically advantageous but less profitable options from the decision set (Mikians et al. 2012). Fleder and Hosanagar (2009) further illustrate how recommendation biases in online platforms systematically narrow consumer choices toward "blockbuster" or profitable items, effectively reducing exposure to diverse and potentially more beneficial products. Consumers often remain unaware of such manipulative anchoring, persistently relying on skewed reference points as if they were objective standards. To address this, policies advocating for transparency in recommendation logic and explicit disclosures of profit-oriented biases could help consumers identify and mitigate detrimental anchoring effects.

4.3 | Algorithm Accuracy, Generative AI, and Overdependence

The rapid proliferation of generative AI has heightened concerns about the accuracy of recommendations and consumer overdependence on these systems, particularly when recommendations are objectively inferior or incorrect (Banker and Khetani 2019). This problem becomes particularly acute when generative AI, which is increasingly powering recommendations, confidently produces suggestions lacking substantive validation or factual accuracy. Overdependence occurs as consumers trust these recommendations without scrutiny, assuming algorithmic outputs are inherently reliable or superior. Policymakers should encourage measures that promote algorithm accountability, including routine accuracy audits, explicit confidence indicators accompanying recommendations, and educational interventions to enhance consumer skepticism and critical evaluation skills. Such steps would reduce the risk of consumer harm from inaccurate yet confidently presented generative AI recommendations.

5 | The Mechanics of Algorithmic Bounded Rationality: A Conceptual Diagram

Figure 1 shows a step-by-step flow to illustrate how consumer biases—especially confirmation bias—are both produced and reproduced by algorithms. In this model, Steps 3 (Data Collection and Algorithmic Learning) and 4 (Personalized Recommendation and Exposure) highlight the core functions of recommender systems, drawing on academic data. Data Collection and Algorithmic Learning (Step 3) uses detailed user histories (clicks, watch times) and metadata to refine predictions (Covington et al. 2016). Personalized Recommendations (Step 4) highlight items deemed most relevant, often magnifying existing preferences (Ekstrand et al. 2018). Engaging repeatedly with similar content consolidates a user's bias (Pariser 2011).

6 | Potential Solutions and Policy Directions

Figure 2 summarizes three complementary approaches: transparency and literacy (Solution A); forced diversity (Solution B), and consumer-centered recommendation agents that re-rank platform lists or operate independently (Solution C). As shown in Potential Solution A in Figure 2, several scholars argue for increased transparency around recommendation engine operation (Brauneis and Goodman 2018). While intricate, deep learning details may remain proprietary, platforms can explain which factors (e.g., watch time and click-through rates) drive recommendations. In addition, consumer education can foster user agency (Hosanagar 2019). Providing tutorials or disclaimers that clarify personalized ranking may encourage individuals to seek broader perspectives or opt out of hyper-personalized feeds.

Potential Solution B in Figure 2 shows multiple possible solutions. Platforms increasingly implement explicit diversity mechanisms to counteract recommendation biases. Netflix's "Play Something" randomly suggests content outside typical viewing patterns, and TikTok's "Refresh" button temporarily resets user preferences, promoting content diversity. Similarly, Amazon's "New to You" badge highlights previously unseen products, and TripAdvisor periodically incorporates less prominent hotel options in search results. These interventions illustrate practical strategies for mitigating algorithmic confinement while maintaining consumer satisfaction. Forced diversity mechanisms disrupt filter bubbles by introducing users to alternative viewpoints or novel products. As shown in Potential

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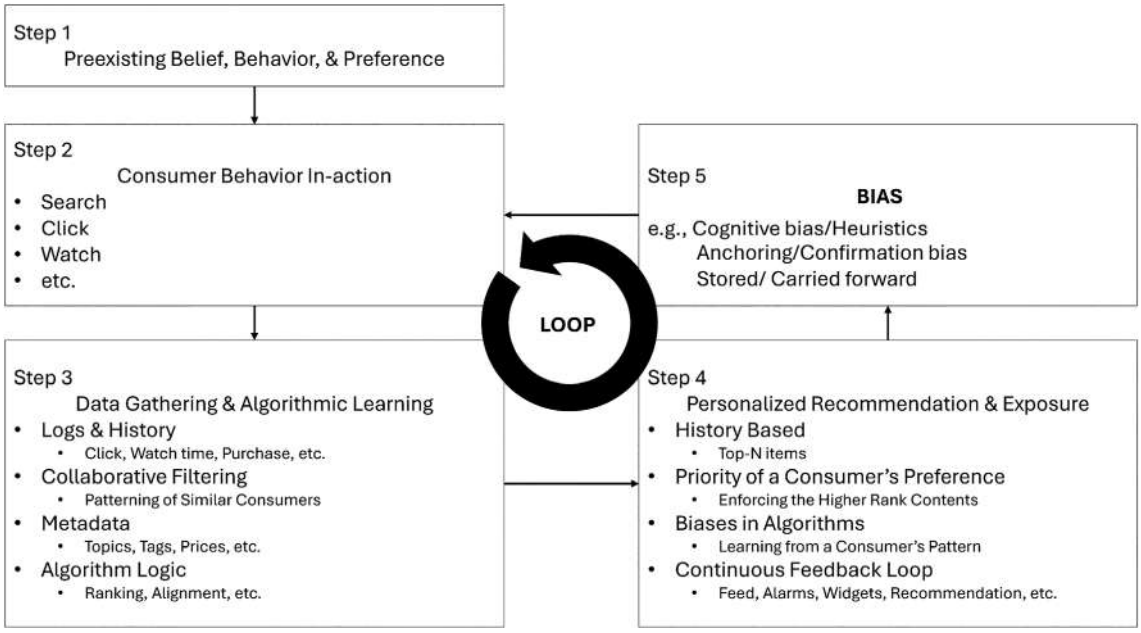


FIGURE 1 | Consumer bias loop by algorithmic recommendation.

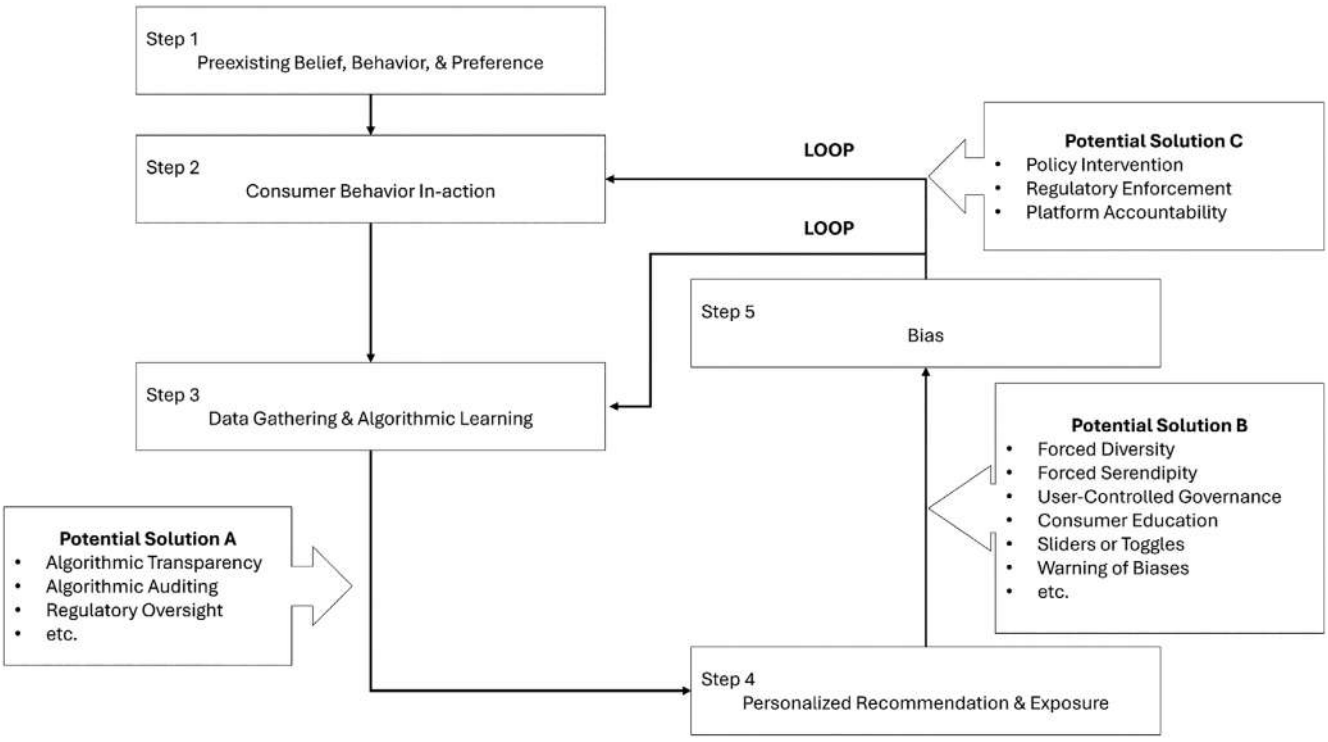


FIGURE 2 | Potential solutions to break the biases loop.

Solution C, user-controlled governance can extend beyond on-platform settings to consumer-centered recommendation agents (user-owned re-rankers operating locally or independently). As algorithmic influence grows, policy should pair platform accountability with data portability and minimal interoperability, so lawful local re-ranking is feasible. Regulators can require periodic impact assessments of bias and content diversity (Brauneis and Goodman 2018), mandate remedies for harms such as deceptive pricing, and support audits and model registries covering both platform and user-side agents. Effective oversight depends on collaboration among platforms, researchers, policymakers, and advocacy groups, with secure data sharing and user-centric standards that prioritize autonomy and welfare (Noble 2018).

7 | Boundary Conditions: When Are Bias Loops Harmful, Benign, or Beneficial?

Much of the existing literature advocates for diversification in recommender systems, primarily to mitigate filter bubble

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TABLE 1 | Risk-exploration matrix.

Domain example	Risk exposure	Exploration need	Bias impact	Recommended solution
Disposable products	Low	Low	Benign	Minimal intervention
Music/film streaming	Low	High	Beneficial	Forced diversity
Insurance or airfare selection	High	Low	Harmful	Transparency, regulatory oversight
Health or investment information	High	High	Mixed	Diversity and confidence indicators

concerns initially raised by Pariser (2011). However, recent reviews (e.g., Bandy 2021; Urman et al. 2024) highlight an insufficiently addressed question: whether recommendation biases are uniformly harmful and if all domains equally require diversity. To address this, the current article introduces a risk-exploration matrix (Table 1), categorizing recommendation contexts along two dimensions: risk exposure (consumer harm resulting from poor recommendations) and exploration need (consumer satisfaction derived from novel content). Risk exposure varies significantly, with minimal risk in routine products versus significant financial impacts in insurance or investments (Hannak et al. 2014). Exploration needs also differ, being critical for entertainment but less important for habitual purchases.

The implications of recommendation systems vary significantly by context. In low-risk, low-exploration scenarios (e.g., disposable products), repetitive recommendations enhance convenience and satisfaction, and forced diversity may unnecessarily complicate user experience (Alba and Hutchinson 1987). *Disposable products* denotes low-cost, high-substitutability goods where errors are inexpensive and easily reversible, so biased repetition primarily compresses search costs and is often functionally benign. Conversely, variety restriction reduces consumer enjoyment in low-risk, high-exploration areas (e.g., music or film streaming), making diversity-enhancing features beneficial (Fleder and Hosanagar 2009). In high-risk, low-exploration contexts (e.g., insurance or airfare selection), biases like price anchoring can cause significant economic harm, necessitating regulatory measures, transparency, and audits (Angwin and Mattu 2016; Hannak et al. 2014). In high-risk, high-exploration contexts (e.g., healthcare or investments), biased recommendations pose significant risks of misinformation, necessitating solutions that combine forced diversity with explicit confidence indicators (Banker and Khetani 2019). In health settings, 'similarity' in recommender embeddings reflects co-consumption patterns rather than clinical equivalence or patient fit, so narrowing can obscure contraindications, insurance/network mismatch, provider quality variation, and wait-time constraints, delaying appropriate care. For queries like "a similar dentist" or "a similar medicine," calibrated diversity plus outcomes- and credential-based filters and explicit confidence/"not medical advice" indicators are warranted. Additionally, discrepancies between perceived and actual recommendation diversity highlight the importance of transparency tools to clarify perceived confinement (Banker and Khetani 2019). Overall, this typology emphasizes tailored, domain-specific solutions rather than one-size-fits-all approaches.

8 | Discussion: Balancing Personalization and Autonomy

Personalization is not inherently problematic, and many consumers value its convenience. The challenge arises when personalization constrains users' knowledge and choices (Hosanagar 2019). Ongoing refinements in predictive algorithms intensify the need to ensure that personalization does not erode autonomy. Addressing this issue requires an interdisciplinary approach that combines behavioral economics, data science, psychology, and policy studies. Ethical principles such as informed consent and meaningful choice must guide any policy or design intervention, ensuring that users do not unknowingly yield to a curated digital sphere. Future studies should combine perceived-constraint scales (Pei et al. 2021) with log-level coverage metrics used in recent audits (Chaney et al. 2018) to quantify the magnitude of Algorithmic Bounded Rationality and evaluate forced-diversity sliders proposed by Banker and Khetani (2019). Personalization need not be firm-centric. Agentic personalization—users deploying their own recommenders—rebalances power, clarifies objectives, and reduces over-reliance on opaque platform metrics. This reframes autonomy as a design primitive rather than an afterthought.

9 | Conclusion

Algorithms significantly shape consumer behavior, amplifying cognitive biases such as confirmation bias, anchoring, and framing, thereby intensifying bounded rationality. These effects raise critical concerns regarding political polarization, consumer welfare, and marketplace competition. Addressing these issues requires several strategies: (a) increasing transparency around algorithm logic, (b) implementing forced diversity mechanisms, (c) enabling user-controlled personalization, (d) establishing regulatory oversight, and (e) fostering stakeholder collaboration. Collectively, these approaches balance personalized recommendations with consumer autonomy, ensuring recommendation systems empower rather than restrict users. Audit-based methodologies, summarized by Bandy (2021) and Urman et al. (2024), provide replicable templates for such evaluations. In addition to transparency, diversity mechanisms, and regulation, consumer-centered

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recommenders offer a practical, low-cost path to align algorithmic curation with user welfare.

Author Contributions

Wookjae Heo: conceptualization, investigation, validation, visualization, writing – original draft, writing – review and editing.

Conflicts of Interest

The author declares no conflicts of interest.

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